

Power Electronics and Closed Loop Current Control for the Galway Energy Efficient Car

Conor Farrell



OLLSCOIL NA GAILLIMHE
UNIVERSITY OF GALWAY

Masters of Electronic & Computer Engineering

Supervisor - Prof. Maeve Duffy
Co-Assessor - Prof. Edward Jones



Abstract

This project introduces a closed-loop current control system for the Galway Energy Efficient Car (Geec), in order to improve efficiency and create repeatable and improved accelerations.

This design is a buck-based brushed DC motor controller. It uses a high-frequency current sensor to implement cycle-by-cycle peak current limiting with a variable, latching limit, enabling dynamic regulation of motor current.

The hardware is divided between two PCBs, the power electronics board primarily a buck converter and the power controller board an isolated control board allowing communication with further devices in the vehicle.

PCB Design

Power Electronics Board

The power electronics board is a 4 layer PCB with several key elements:

Press Fit Connector: High power M5 screw connector for ring terminals rated for 60–80A, chosen for its low-resistance.

Digital Joulemeter: I²C joulemeter with an integrated shunt resistor, allowing for immediate and constant measurement of energy, power current etc.

Functional Isolation: Using both digital and analog isolators to create a functional isolation between the microcontroller and the primary power electronics.

Indicator LED & Test Points: Optional LEDs to indicate power integrity and the activity of PWM and startup, important signals expose a testpoint

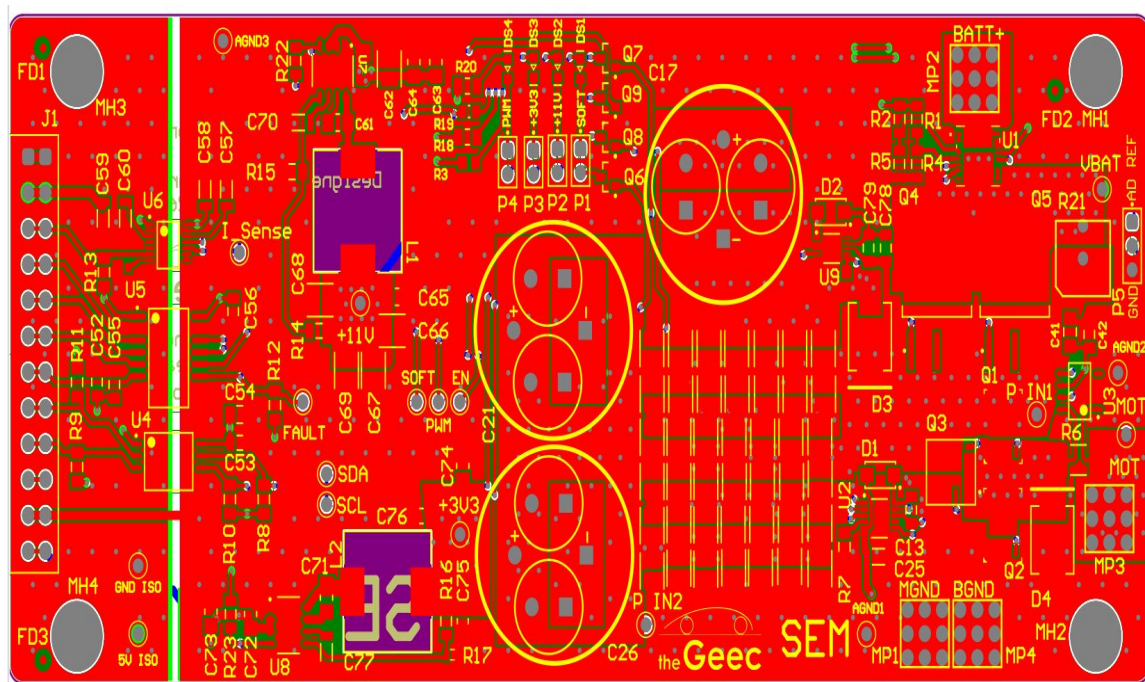


Figure 1: Power Electronics Stackup

Power Controller Board

The power controller primarily handles the implementation of hardware-based current control, and it also integrates:

CAN: This allows for messages to the DAQ and Steering Wheel, improving data analysis, feedback from sensors and improvements to driver information.

Motor Voltage Sensor: Another isolated amplifier with a low pass filter allows for an accurate voltage at the motor terminals.

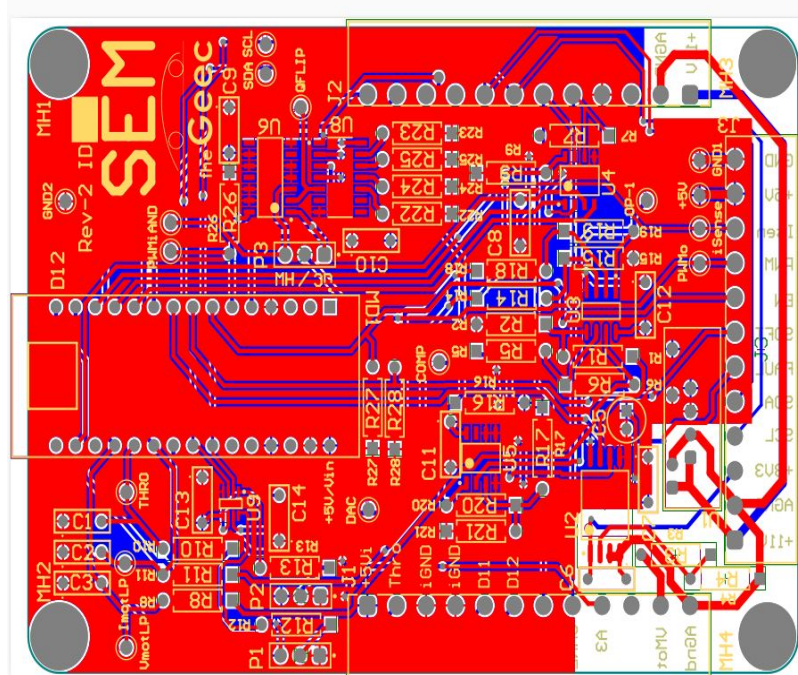


Figure 2: Power Controller Stackup.

Finished PCBs

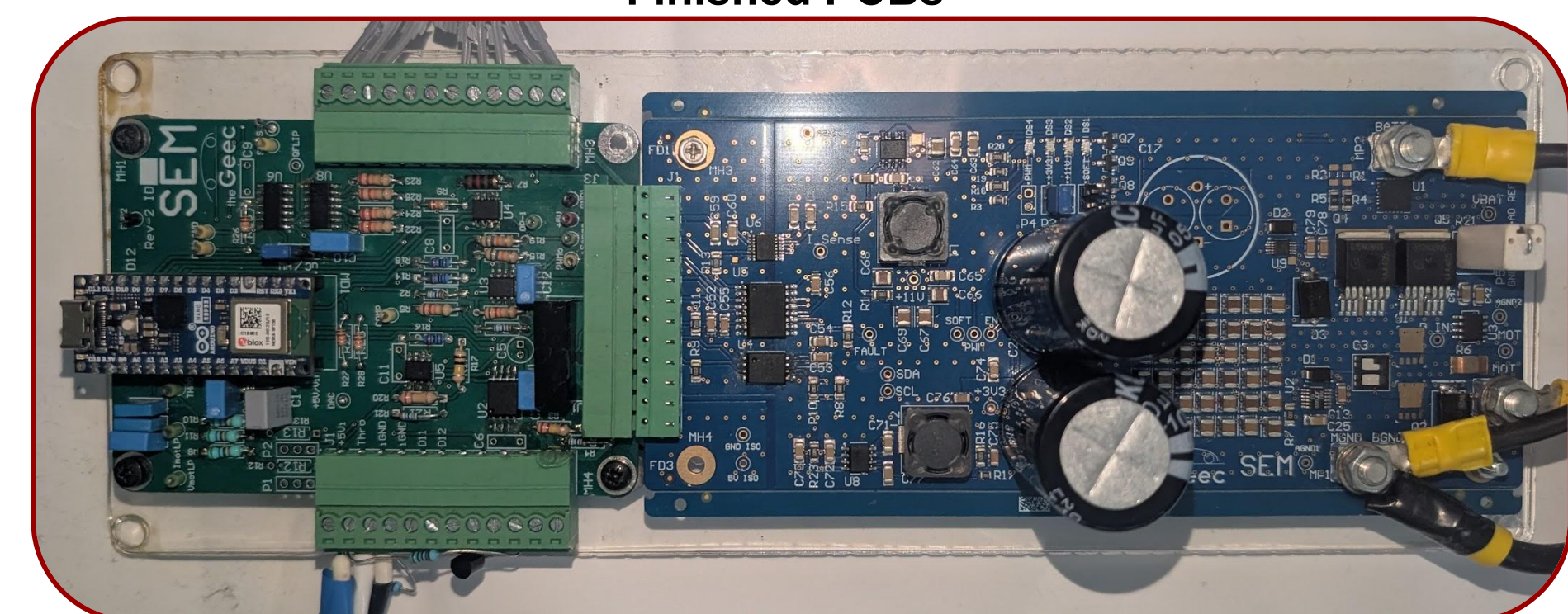


Figure 3: Finished PCBs

Half-Bridge MOSFET: Infineon MOSFET that integrates both the high and low side MOSFET into one IC.

Soft Start: 10Ω resistor to charge capacitor bank, with parallel MOSFETs.

Hardware Current Control

The Geec previously used open loop control with a fixed ramp rate to limit peak current. This relied on the driver to monitor the acceleration and caused high peak currents during acceleration.

The new design uses a variable limit with a latching function to implement cycle-by-cycle peak current limiting.

In real terms this allows for the removal of the ramp rate for acceleration, and relies on the peak current limit instead. The DC motor directly converts current to torque based on:

$$Nm = K_t I$$

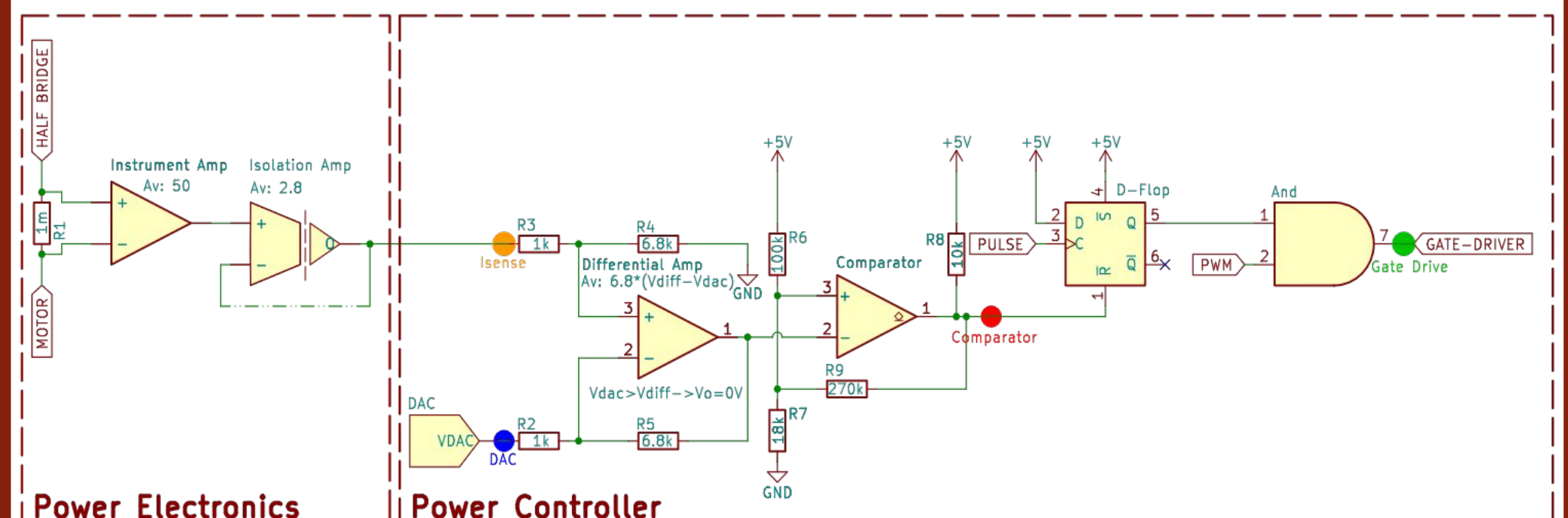


Figure 4: Current Control Schematic.

Working Principle

Instrument Amplifier: A high bandwidth current amplifier over a small resistor.

Comparator: Uses a fixed hysteresis with V_{TL} : 1V and V_{TH} : 0.7V.

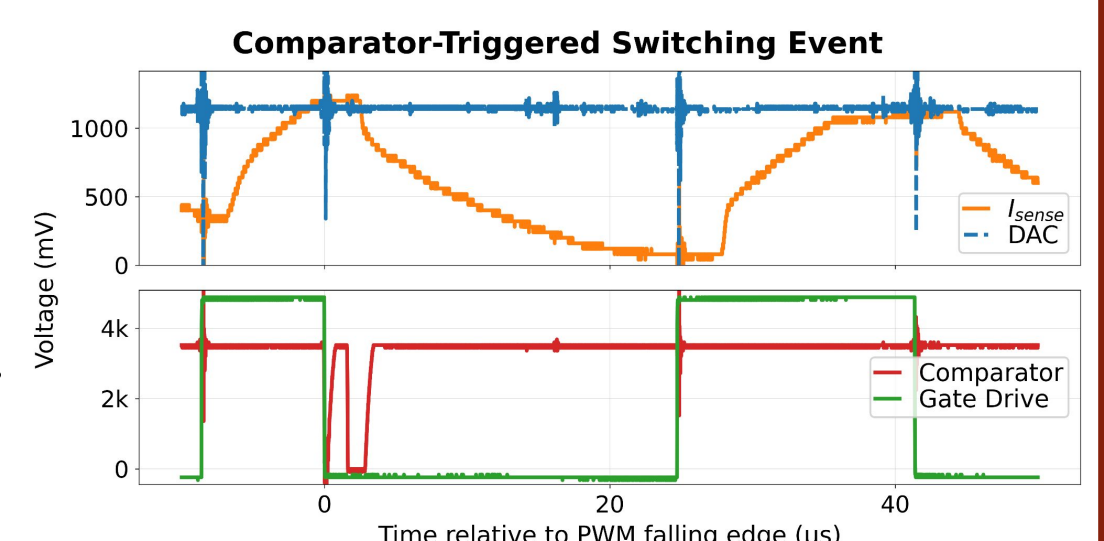
DAC: The DAC can provide an output voltage up to 4V.

Differential Amplifier: The rail-to-rail Op Amp, outputs 0V when DAC > Isense, and a gain of 6.8 on the difference between Isense and DAC.

Flip Flop: The comparator pulls Reset low, Output Q is latched low, on the next clock cycle Q returns high, D is tied high.

Conclusion

On the race track the Geec will have to accelerate in a on flat ground, up hills and around corners. The fixed limit will regulate acceleration, even in dynamic conditions.



Results

The results to the right were collected with the dynamometer repeating multiple accelerations at the same approximate torque ~5Nm and trying to accelerate and decelerate to the same speeds.

Figure 6 emulates the old open loop system vs the new current controlled setup. With constant load acceleration, current control results in a 10% improvement over the 5s open loop ramp.

Figure 7 shows the effect of increasing the switching frequency and also increasing the current limits. Further investigation can be done to determine optimal configuration of frequency and current limit.

Energy per Acceleration Run: 15A Limit, Varying Ramp Duration

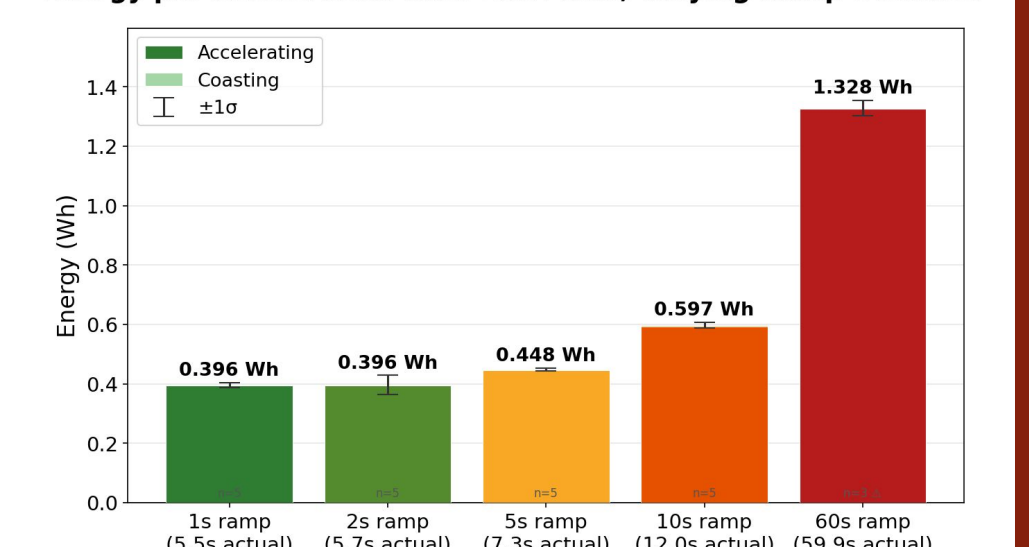


Figure 6: Energy Consumption vs Ramp Rate
Energy Comparison: Varying Switching Frequency & Current Limit

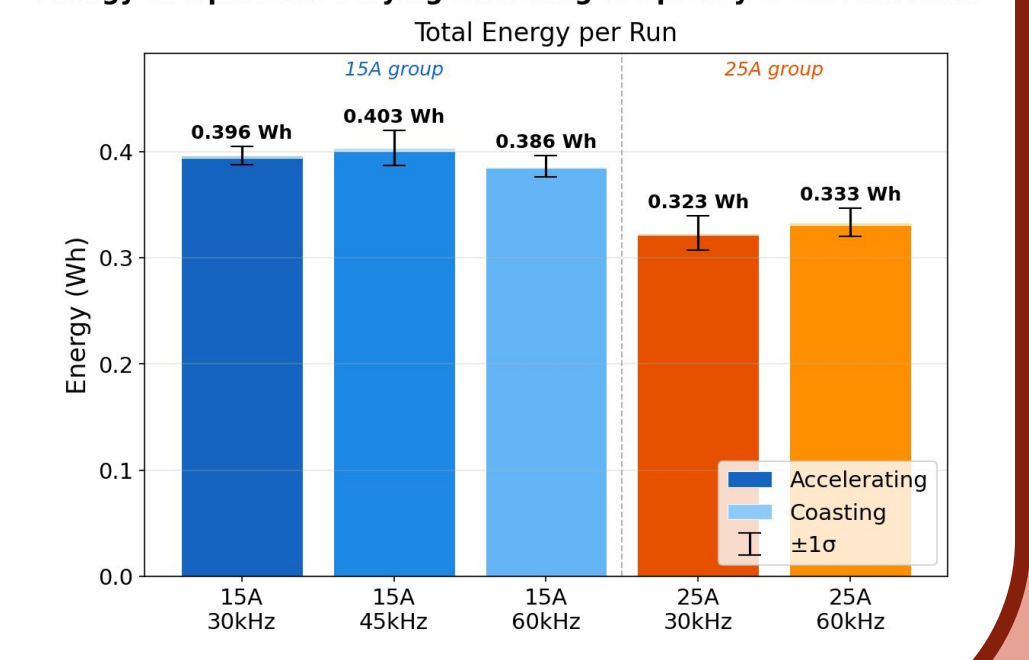


Figure 7: Energy Consumption vs current limit

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